

An acknowledgement is invalid if the acknowledged sequence number does not correspond to an outstanding, unacknowledged BTP packet sequence number. In contrast to TCP, BTP acks are not "free." A stand-alone ack—that is, a BTP packet that contains a packet receipt acknowledgement value but no buffer segment payload—consumes a slot in a remote peer's window just like any other packet. Stand-alone acknowledgement packets SHALL be acknowledged by a remote peer. The implications of this are examined in [Section 4.19.4.9, "Idle Connection State"](#).

Each peer SHALL maintain an acknowledgement-received timer. When a peer sends any BTP packet, it SHALL start this timer if it is not already running. The timer's duration SHALL be globally defined as BTP_ACK_TIMEOUT seconds, referred to as the acknowledgement timeout interval.

A peer SHALL restart its acknowledgement-received timer when a valid acknowledgement is received for any but its most recently sent unacknowledged packet. A peer SHALL stop its acknowledgement-received timer if it receives an acknowledgement for its most recently sent unacknowledged packet. If a peer's acknowledgement-received timer expires, or if a peer receives an invalid acknowledgement, the peer SHALL close the BTP session and report an error to the application.

Because the server's handshake response bears an implicit BTP sequence number of zero, a server SHALL start its acknowledgement-received timer when it sends a handshake response.

Each peer SHALL also maintain a send-acknowledgement timer. When it receives any BTP packet, a peer SHALL record the packet's sequence number as the corresponding BTP session's pending acknowledgement value and start the send-acknowledgement timer if it is not already running. The timer's duration SHALL be defined as any value less than one-half the acknowledgement timeout interval. This ensures that on a healthy BLE connection, a peer will always receive acknowledgements for sent packets before its acknowledgement-received timer expires.

A peer SHALL stop its send-acknowledgement timer when any pending acknowledgement is sent, either as a stand-alone BTP packet or piggybacked onto an outgoing buffer segment. If this timer expires and the peer has a pending acknowledgement, the peer SHALL immediately send that acknowledgement. If the peer sends any packet before this timer expires, it SHALL piggyback any pending acknowledgement on the transmitted packet and stop the send-acknowledgement timer.

Because the server's handshake response bears an implicit BTP sequence number of zero, a client SHALL set its pending acknowledgement value to zero and start its send-acknowledgement timer when it receives the server's a handshake response. Operation of the send-acknowledgement and acknowledgement-received timers is illustrated in [Figure 26, "BTP session lifecycle for Central acting as GATT Client"](#) in [Section 4.19.4.11, "Protocol State Diagrams"](#).

If a peer detects that its receive window has shrunk to two or fewer free slots, it SHALL immediately send any pending acknowledgement as a stand-alone BTP packet. This prevents the session from stalling in the interval between when a peer's receive window becomes empty and when its send-acknowledgement timer would normally fire.

4.19.4.9. Idle Connection State

When neither side of a BTP session has data to send, BTP packets will still be exchanged every send-acknowledgement interval due to acknowledgements generated by the receipt of previous data or stand-alone acknowledgement packets, as discussed in [Section 4.19.4.8, "Packet Acknowledgements"](#). The behavior of the acknowledgement-received timer in this scenario doubles as a keep-